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ORIGINAL RESEARCH

Power frequency magnetic field interference suppression method for online frequency response analysis of power transformers

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Abstract

Frequency response analysis is widely used for the offline diagnosis of winding deformations in power transformers. To apply this approach to a working transformer, the magnitude of the response current needs to be measured by using Rogowski coil sensors across a load current. The saturation of the power frequency magnetic field in these current sensors must be prevented to ensure accurate measurement of such small response currents. Here, a method is presented to suppress the power frequency magnetic field using a sensing system including a special connection of three-phase current sensors based on the sum of the three-phase power frequency load currents of the transformer being close to zero. Each sensor comprises two secondary coils: a measuring coil and an anti-saturation coil. The anti-saturation coils are connected in parallel with one another through small inductors to eliminate the power frequency magnetic field in the cores of the sensors. Theoretical analysis is used to derive a solution for this system. The experimental results verify the proposed method as enabling the sensors to function with a transformer carrying a load current of 2333 A.

1 | INTRODUCTION

Transformers are among the most important electrical equipment used in power networks, and their safe operation is crucial for that of power systems [1]. Thus, defects in transformers need to be identified before they lead to accidents or failures [2]. Compared with new transformers, older transformers are more likely to encounter winding deformation defects due to deficiencies in their design, short-circuit resistance calculations, manufacturing and material selection. When the transformer is subjected to a short-circuit current, its windings may be deformed because they cannot withstand the resulting electromagnetic force. A minor winding deformation can gradually develop into a large deformation after experiencing multiple instances of large currents [3], and deformed windings are significantly mechanically weaker than normal windings. Transformers with deformed windings are thus more likely to be damaged. According to the International Council on Large Electric Systems (CIGRE), one-third of all transformer failures are caused by winding deformations [4].

Many methods are available for the offline detection of winding deformations, and frequency response analysis (FRA) is the most widely used method in the power industry due to its high sensitivity and accuracy [5, 6]. The Chinese power industry and the International Electrotechnical Commission (IEC) have proposed standards for the application of FRA [7, 8].

However, powering off a transformer is problematic because of the high demand for continuous power supply. There is an urgent need for the FRA method to be applied to transformers during operation. Notable studies have thus focused on the problem that needs to be solved first, that is, the means by which signals are injected into a live winding and the means to obtain response signals. In 2009, De Rybel proposed using the voltage tap of the secondary screen of the bushing or the

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lead wire of the end screen to inject a sweep frequency signal into the windings of an operating transformer, followed by obtaining the response signal from the lead wire of the end screen of the bushing [9]. Additionally, a circuit for suppressing power frequency voltage interference in this method was provided in ref. [10] in 2012. In 2011, Hossein Borsi and Ernst Gockenbach of Hannover University in Germany proposed a non-invasive capacitive sensor. The capacitance between the sensor and the high-voltage conductor in the bushing can be used to inject an excitation signal into the winding and to measure the response voltage signal by electric field coupling [11]. This capacitive sensor method was developed and applied to an offline transformer with a rated voltage of 110 kV in 2014 [12] and 2017 [13]. Moreover, methods based on the capacitance of the end screen of the bushing or non-invasive capacitive sensor have undergone limited improvement in recent years. We believe that this limitation is because substation personnel resist installing devices on the end screen of the bushing or pasting metal sheets on the surfaces of the bushings to ensure the safety of the transformer. In addition, relatively greater attention has been given to the application of artificial intelligence in FRA diagnosis [14–16].

We previously proposed a method for injecting an excitation signal into a working transformer and measuring the response current via a Rogowski coil sensor in 2016 [17] and confirmed the safety of this method for a working 35 kV power transformer in 2017 [18]. However, the windings of transformers bear power frequency load current on the order of hundreds or even thousands of amperes. This current usually causes the magnetic core of the response current sensor used for FRA to become saturated, preventing it from measuring FRA signals. In previously reported experiments, we measured the FRA signals at a substation by using a coil-type current sensor. The FRA signal could be measured only when the power frequency current crossed zero [18]. This is clearly not a reliable solution.

In this paper, we present a method to suppress the interference caused by the power frequency load current during measurements of the current signals for FRA, particularly to supress the magnetic field in the core of the sensors. We provide the basic principle and a theoretical analysis of the proposed method, and verified its feasibility and accuracy through experimental evaluations.

2 | PROBLEM OF MAGNETIC FIELD SATURATION IN A CURRENT SENSOR DURING AN ONLINE MEASUREMENT

2.1 | Application scenario of current sensors for online FRA

Frequency response analysis is widely used for the offline diagnosis of winding deformations in power transformers. This approach includes considering the windings of the transformer as an inductance–capacitance network. The deformation of the windings leads to changes in the inductance and capacitance of the network and, thus, in the transfer function of the network. For example, as shown in Figure 1, a swept frequency voltage U_i is applied to the neutral terminal of a power transformer using

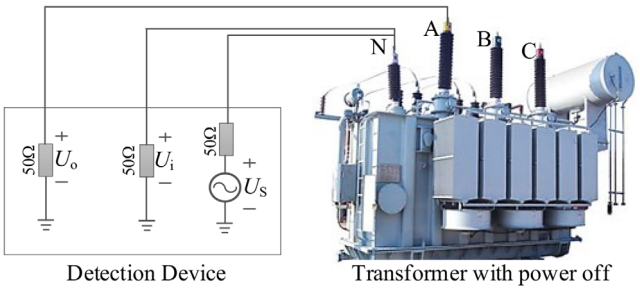


FIGURE 1 Measurement for the offline FRA of power transformers. A-A phase, B-B phase, C-C phase, N-Neutral point.

a source of voltage U_S, and the response voltage U_o is measured at the high-voltage terminal of the A-phase winding. The amplitude of the transfer function H of the A-phase winding is calculated as U_o/U_i. If H is different from its original value, the windings have been mechanically deformed. The frequency of U_i usually ranges from 1 kHz to 1 MHz.

This principle can be extended to online FRA. For example, the excitation signal is injected into and the response signal is measured on a working transformer. Furthermore, the transfer function of the windings is calculated by using the signals to diagnose winding deformations in the transformer.

Of note, there are differences between offline and online FRA. Because live windings are at a high electric potential, it is difficult to measure the voltage type of response signals on windings. However, current sensors based on a Rogowski coil can be installed outside the high-voltage bushing to measure the response current on the windings. Therefore, online FRA is based on the current type of response signal.

A typical scenario for the application of current sensors for online FRA to an SSZ11-50000 kVA/110 kV power transformer is shown in Figure 2. This scenario is used for the present study. When the excitation current I_S is injected into the neutral point N of the windings through a coil (called the excitation coil) by a voltage source U_S, an induced potential U_N appeared at the neutral point, and response currents I_{AF1}, I_{BF1}, and I_{CF1} appeared in the high-voltage terminal of the three-phase windings. The sensors are installed on the oil tank of the transformer (around the high-voltage bushings) to measure the response current.

Therefore, instead of the voltage type of the response signal, the current type of the response signal on the windings is used to execute FRA on a transformer that is in operation. According to our previous studies, the FRA current on the high-voltage terminals of windings has a very small magnitude, ranging from tens of microamperes to several milliamperes. However, there are hundreds or thousands of amperes of power frequency load current in the windings. We must find a way to sense milliamps of FRA current under the interference of hundreds of amperes of power frequency current.

2.2 | Requirement for the current sensor for FRA

In order to avoid typographic problems, phasors such as voltage, current, magnetic field, and electric potential are rep-

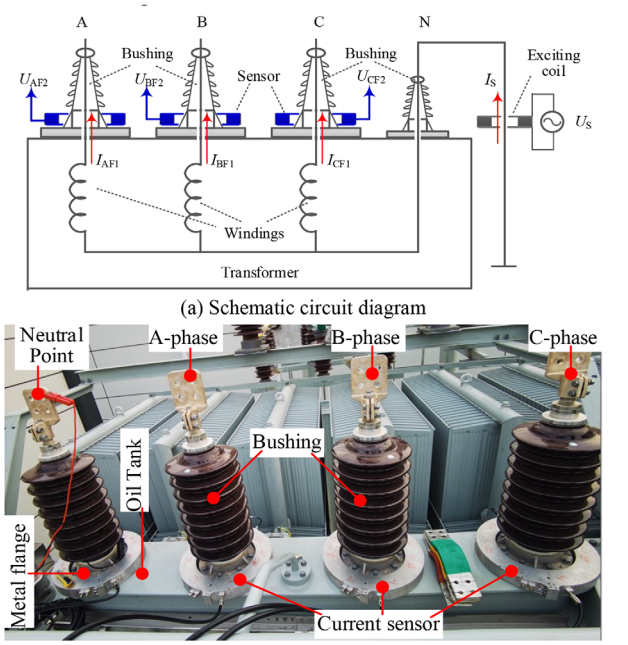


FIGURE 2 Scenario for the application of the current sensors for online FRA on power transformers.

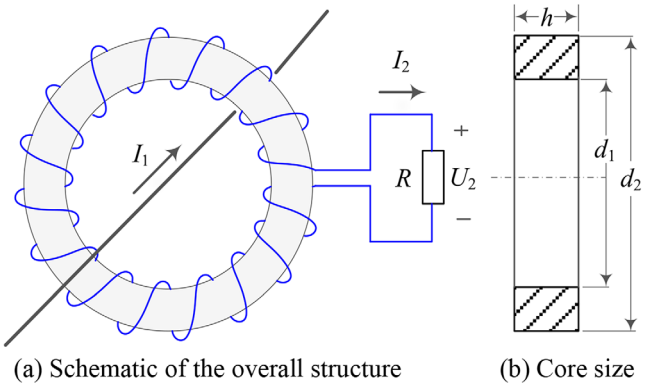


FIGURE 3 Theory and structure of the coil sensor.

resented by common letters. The coil-type current sensor converts the measured current I_1 into the output voltage U_2. The theory and structure of the coil-type sensor are shown in Figure 3.

Considering the frequency of the current I_1 and the value of the integral resistance R in our application scenario, the capacitance and resistance of the wire are ignored. Therefore, when the cross section of the core is rectangular, the following formula exists [19]:

$$j\omega MI_1 = j\omega LI_2 + U_2 = j\omega LI_2 + RI_2 \tag{1}$$

$$L = Mn = \frac{\mu_0\mu_r hn^2}{2\pi}\ln\frac{d_2}{d_1} \tag{2}$$

$$Z_C(j\omega) = \frac{U_2}{I_1} = \frac{j\omega MR}{j\omega L + R} \tag{3}$$

$$f_L = \frac{1}{2\pi}\frac{R}{L} \tag{4}$$

where j is the sign of the imaginary part of the variable. n is the number of turns of the coil. L is its inductance. M is the mutual inductance between the coil and the straight conductor bearing the measured current I_1, and I_1 is surrounded by the coil. I_2 is the current in the coil, and R is its integral resistance. U_2 is the output of the sensor. ω is the angular frequency of the signal, μ_r is the relative magnetic permeability of the material of the core of the coil, μ_0 is the permeability in a vacuum, and h, d_1, and d_2 are the height, inner diameter, and outer diameter of the core of the coil, respectively. Z_C is the transfer impedance of the sensor, which describes the operating characteristic of the sensor. f_L is the lower cut-off frequency of Z_C.

It is a loose coupling between the straight conductor and the coil. And the coupling coefficient is not equal or even close to 1. However Equations (1)–(4) have universal meaning regardless the value of the coupling coefficient, and following equations were infered from (1)–(4).

The current sensor for FRA requires a wide frequency band ranging from 1 kHz to 1 MHz, high sensitivity (that is, high Z_C), and an open-closed structure that is, suitable for field installation. The first feature of the current sensor for online FRA is the large diameter and small height of the sensor because of the restricted space around the boot of the bushings. As shown in Figure 2, the inner diameter of the sensor needed to be sufficiently large to surround the bushing, and its height needed to be smaller than that of the metal flange of the bushing (less than 50 mm). According to (1)–(4), using a soft magnetic core with a large μ_r is the only approach to obtain an f_L of 1 kHz while maintaining a large Z_C for a sensor with such a large diameter and limited cross-sectional area.

For example, we designed a core made of an iron-based amorphous material with a d_1 of 290 mm, d_2 of 350 mm, and h of 25 mm to construct a sensor for 35 kV bushings of transformers. The equivalent μ_r of the core at 50 Hz was approximately 3000, although the material of the core had a permeability above 50,000 because the open closed structure yielded a cutting gap in the core. We obtained the best characteristics of the sensor at $n = 150$ and $R = 330\ \Omega$, as shown in Figure 4 (the blue curve). If the magnetic permeability of the core is not sufficiently large (for example, if the core has a gap of 2 mm), then the f_L is much greater than 1 kHz, therefore the sensor will have worse sensitivity to the signals at 1 kHz, as shown in Figure 4 (the red curve).

2.3 | Magnetic saturation of the current sensor

The above analysis and experiments showed that the magnetic core is the key component for ensuring the desired measurement band of the sensors. However, the power frequency load currents (called I_{AP1}, I_{BP1}, and I_{CP1} for the three phases and generally referred to as I_{P1}) flow in the three wind-

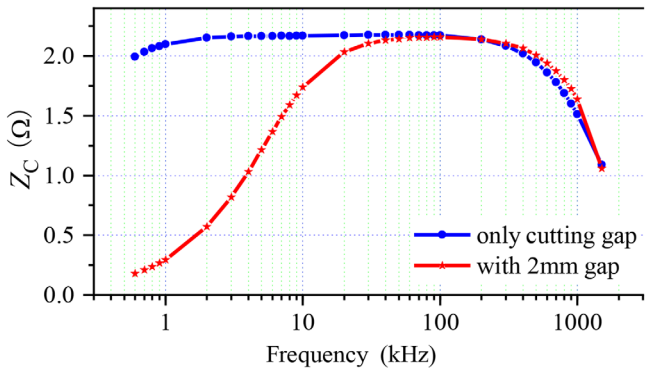


FIGURE 4 Transfer characteristics of the coil type current sensors.

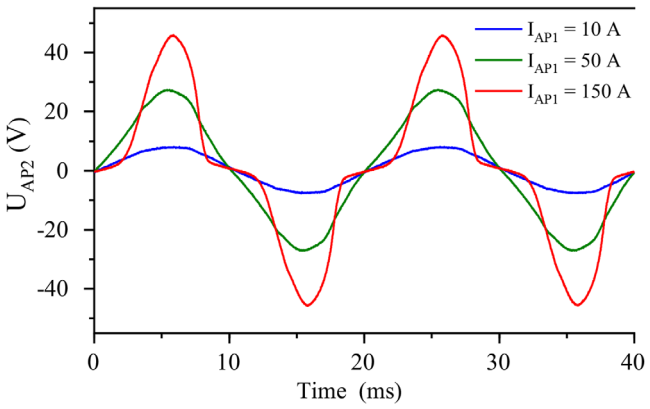


FIGURE 5 Output of the single sensor to the power frequency current.

ings in a working transformer. For the common SSZ11-50000 kVA/110 kV-type transformer, the rated values of I_{AP1} with 110, 35, and 10 kV windings were 262.4, 789.3, and 2749.3 A, respectively. Such a large current inevitably leads to saturation of the magnetic field at the core of the sensor.

The output voltage U_{AP2} of the sensor in response to a 50 Hz power current I_{AP1} is shown in Figure 5. When I_{AP1} was 50 A, the waveform of U_{AP2} deviated slightly from the sinusoidal waveform, and its peak value was 28 V. When I_{AP1} was 150 A, the waveform of U_{AP2} was strongly distorted. Specifically, the critical saturation current of the sensor was 50 A. Therefore, the normal load current of the transformer can cause the core of the sensor to become deeply magnetically saturated. This effect can lead to severe magnetostrictive vibrations that cause the sensor to fail. Furthermore, a large level of power frequency interference in the output of the sensor (e.g., peak value of 45 V when I_{AP1} was 150 A) leads to severe difficulties in detecting small FRA current signals.

2.4 | Conventional methods for suppressing the saturation of a magnetic field at the core of a current sensor

Two types of methods are commonly used to suppress the saturation of the magnetic field in the core of a coil-type current sensor under the influence of a large power frequency current.

The first method includes reducing the magnetic permeability μ_r of the core by using a material with a low permeability or equivalent by creating a gap in the path of the loop of the core. The disadvantage of this method is to increase the f_L of the sensor, especially of the sensor with a gap of 2 mm, as shown in Figure 4. Thus, the frequency band of the sensor does not meet the requirements.

The second method includes using an inductive integral impedance. The resistance R is replaced by the parallel combination of R and an inductance L_C. According to calculations using (1)–(4), if the output voltage U_{AP2} is compressed to 5 V when the 50 Hz power current I_{AP1} is 500 A, the integral inductance L_C needs to be 4.785 mH. However, the f_L of the sensor with this inductor increases beyond 20 kHz, which is much larger than the required 1 kHz. Therefore, it is nearly impossible to suppress the power frequency noise while maintaining the required frequency band using inductive integral impedance.

Thus, conventional methods cannot overcome the magnetic saturation of the current sensor for online FRA, and novel methods are necessary and urgent.

3 | SUPPRESSING THE POWER FREQUENCY MAGNETIC FIELD

The magnetic field induced by the power frequency current significantly increases the difficulty of frequency response current measurement. Below, we present a method for suppressing power frequency magnetic field in the sensor based on three-phase current sensors.

3.1 | Basic structure and principle

The basic principles of coil-type current sensors, electromagnetic induction and Lenz's theorem inform that a fundamental technique for eliminating the magnetic field is to provide a sufficiently large current I_2 in the coil (that is, the secondary-side current). This secondary-side current can offset the magnetic field caused by the measured current I_1 (that is, the primary-side current).

We thus design a novel sensing system that comprises three-phase sensors to provide a sufficiently large secondary-side current for each sensor. Specifically, this effect is maintained only by the power frequency current, not the FRA current. The structure and circuit topology of the sensing system are shown in Figure 6. The three sensors have identical structures and configurations, including their cores and numbers of turns. The output ports of the sensors are connected in parallel with one another and connected to the integral resistance R. The equivalent circuit diagram is shown in Figure 6b,c.

The three-phase windings of the power transformer are represented by the impedances Z_{WA}, Z_{WB}, and Z_{WC}. The power frequency electric potentials are represented by E_A, E_B, and E_C; the three-phase loads of the transformer are represented by

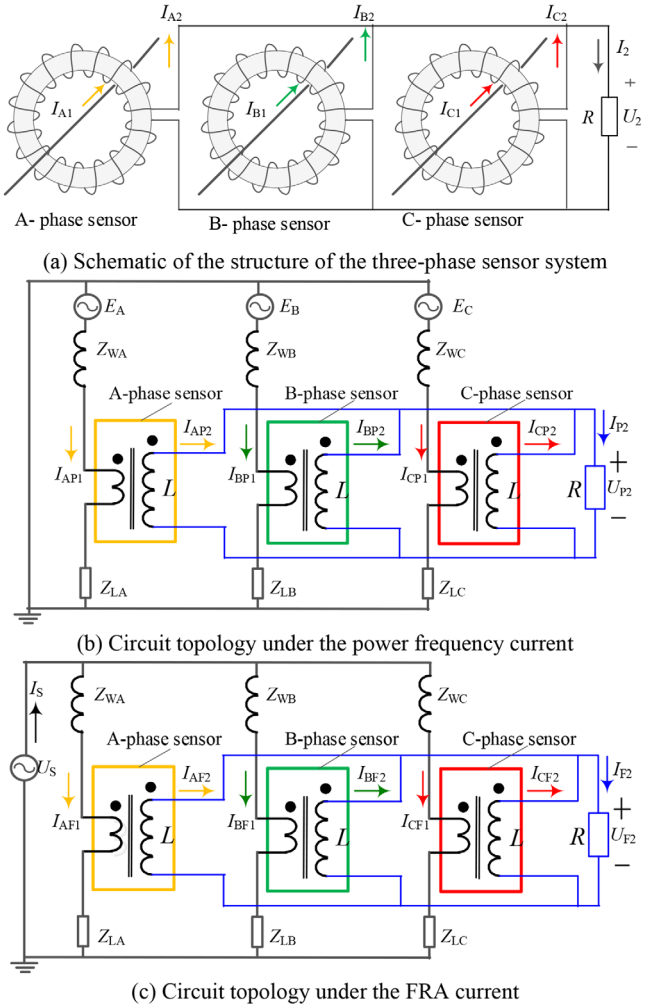


FIGURE 6 Principle of the method for suppressing the power frequency magnetic field based on three-phase sensors.

Z_{LA}, Z_{LB}, and Z_{LC}; and the excitation voltage and the current for FRA are represented by U_S and I_S, respectively.

In the case of a power frequency load current, the three-phase load currents are usually balanced, and we obtain the following:

$$I_{AP1} + I_{BP1} + I_{CP1} \approx 0 \tag{5}$$

Therefore:

$$I_{AP2} + I_{BP2} + I_{CP2} = I_{P2} \approx 0 \tag{6}$$

$$U_{P2} = RI_{P2} \approx 0 \tag{7}$$

By substituting (7) to (1) and then to the expression of the magnetic field strength in the core, we obtain the peak value of the largest magnetic field in the core:

$$B_{mag} = \sqrt{2}\frac{\mu_0\mu_r |I_{AP1} - nI_{AP2}|}{\pi d_1} \approx 0 \tag{8}$$

That is, the power frequency magnetic field in the cores of the sensors is approximately zero such that the problem of magnetic saturation does not occur. In comparison with the conventional method based on integral impedance, the B-phase and C-phase sensors take as the integral impedance Lc of the A-phase sensor in this structure, and vice versa. However, in our proposed sensor system, the integral impedance Lc is an active (rather than passive) component. This impedance includes the induced potential from I_{BP1} and I_{CP1} that neutralizes the induced potential coupled from I_{AP1}. Therefore, this equivalent active integral impedance more effectively suppresses the power frequency magnetic field than does the conventional integral impedance.

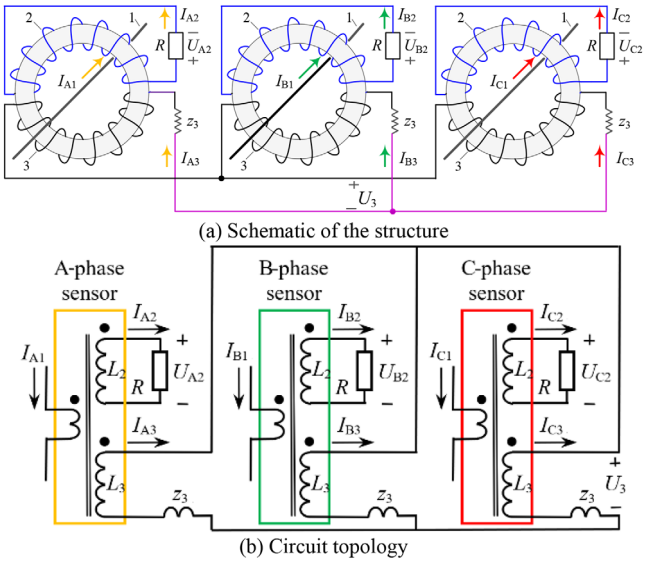


FIGURE 7 Improved method of suppression based on three-phase sensors.

To simplify our description of the currents for FRA, we assume that the impedances Z_{WA}, Z_{WB}, and Z_{WC} are equivalent, as are the impedances Z_{LA}, Z_{LB}, and Z_{LC}. This hypothesis does not affect our basic conclusion. We then obtain the following:

$$U_{F2} = R\left(I_{AF2} + I_{BF2} + I_{CF2}\right) = \frac{R}{n}\left(I_{AF1} + I_{BF1} + I_{CF1}\right) \tag{9}$$

Therefore, the response current for FRA can be accurately measured. The characteristics of the sensing system for FRA current is the same as that of a single sensor.

3.2 | Improved structure and principle

Equation (9) clearly shows that the basic sensing system depicted in Figure 6 cannot distinguish the response current I_{AF1} from I_{BF1} and I_{CF1}. If this basic sensing system is used for online FRA, then the three-phase windings of the power transformer are considered as a whole rather than as three components. The deformation in the windings can be diagnosed but not located at which phase winding occurs.

We thus propose an improved structure to distinguish the response current in each winding. The improved structure and its circuit topology are shown in Figure 7. The three sensors have identical structures and configurations, including the core,

its turns, and its integral resistance. Each sensor has two coils on the same magnetic core. One coil (No. 2) is called the measuring coil, while the other (No. 3) is called the anti-saturation coil. The anti-saturation coils of these three sensors are connected in parallel, and three small impedances z_3 are connected in series at the output port of the anti-saturation coil.

To borrow the expression of the primary side and secondary side of a current transformer, the conductor (coil 1) in the bushing is the primary-side coil, and coils No. 2 and 3 are secondary-side coils. The circuit topology of the three-phase sensors is shown in Figure 7b, where I_{A1}, I_{B1}, and I_{C1} represent the total current on the windings of the three phases; I_{A2}, I_{B2}, and I_{C2} represent the total current on the measuring coils; and I_{A3}, I_{B3}, and I_{C3} represent the total current on the anti-saturation coils. The outputs of the total voltage of coil 2 are U_{A2}, U_{B2}, and U_{C2}, which are also the outputs of the sensing system. Coil 3 has the same output as U_3.

The relationship between coils 1, 2, and 3 can be inferred from (1). Consider the A-phase sensor as an example:

$$RI_{A2} = n_2 \frac{d\varphi_A}{dt} = j\omega k n_2 (I_{A1} - n_2 I_{A2} - n_3 I_{A3}) \quad (10)$$

$$U_3 + z_3 I_{A3} = n_3 \frac{d\varphi_A}{dt} = j\omega k n_3 (I_{A1} - n_2 I_{A2} - n_3 I_{A3}) \quad (11)$$

where n_2 is the number of turns of coil 2, n_3 is that of coil 3, φ_A is the magnetic flux in the core, and k is the coefficient of inductance:

$$k = L_2 / n_2^2 = L_3 / n_3^2 = \frac{\mu_0 \mu_r h}{2\pi} \ln \frac{d_2}{d_1} \quad (12)$$

where L_2 and L_3 are the inductances of coils 2 and 3, respectively. The corresponding formulae also hold for the sensors in phases B and C.

The total current consists of the FRA current and the power frequency current. We thus analyse these two types of currents according to the principle of superposition.

3.3 | Effect on the power frequency current

We consider only the power frequency current in the improved sensor system in this section, that is, $I_{A1} = I_{AP1}$, $I_{A2} = I_{AP2}$, $\omega = \omega_P$, $I_2 = I_{P2}$, $U_{A2} = U_{AP2}$, and $U_3 = U_{P3}$, as well as the B-phase sensor and C-phase sensor.

The sum of the three-phase load currents of the power transformer is approximately zero. The sum of the currents in the three coils No. 3 is zero. Therefore, we obtain the following by combining (10) for the three-phase sensors:

$$R(I_{AP2} + I_{BP2} + I_{CP2}) \approx -j\omega_P k n_2^2 (I_{AP2} + I_{BP2} + I_{CP2}) \quad (13)$$

The condition for satisfying (13) is as follows:

$$I_{AP2} + I_{BP2} + I_{CP2} \approx 0 \quad (14)$$

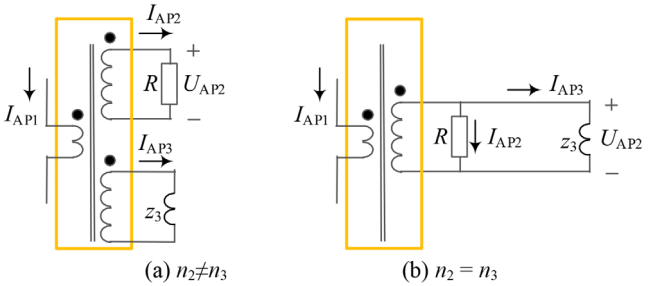


FIGURE 8 Equivalent circuit to the power frequency current.

We combine (11) for the three-phase sensors with (14) to obtain the following:

$$U_{P3} \approx 0 \quad (15)$$

Furthermore, we obtain the following equation as long as z_3 is a small impedance on the order of μH:

$$U_{AP2} = n_2 \frac{d\varphi_{AP}}{dt} \approx \frac{n_2}{n_3} U_{P3} \approx 0 \quad (16)$$

$$I_{AP2} = U_{AP2} / R \approx 0 \quad (17)$$

To understand this phenomenon, an even more simplified circuit topology is shown in Figure 8. When $U_{P3} = 0$, the circuit in Figure 7b is made equivalent to that in Figure 8a, and it is further made equivalent to that in Figure 8b if $n_2 = n_3$. Clearly, the magnetic flux caused by I_{AP1} is essentially neutralized by that caused by I_{AP3}, and $I_{AP2} \approx 0$ because the impedance of z_3 is much less than the value of R.

Therefore, the magnetic field in the cores of the sensors is close to zero because the sum of the three-phase power frequency currents is approximately zero.

3.4 | Effect on FRA current

We consider only the FRA currents in the improved sensing system in this section, that is, $I_{A1} = I_{AF1}$, $I_{A2} = I_{AF2}$, $I_2 = I_{F2}$, $U_{A2} = U_{AF2}$, and $U_3 = U_{F3}$, as well as the B-phase sensor and C-phase sensor.

Considering that the sum of the currents in the three coils No. 3 is zero and combining (10) or (11) for the three-phase sensors, we obtain:

$$I_{AF2} + I_{BF2} + I_{CF2} = \frac{j\omega k n_2}{R + j\omega k n_2^2} (I_{AF1} + I_{BF1} + I_{CF1}) \quad (18)$$

$$3U_{F3} = j\omega k n_3 [I_{AF1} + I_{BF1} + I_{CF1} - n_2 (I_{AF2} + I_{BF2} + I_{CF2})] \quad (19)$$

Therefore:

$$U_{F3} = \frac{j\omega k n_3 R}{3R + 3j\omega k n_2^2} (I_{AF1} + I_{BF1} + I_{CF1}) \quad (20)$$

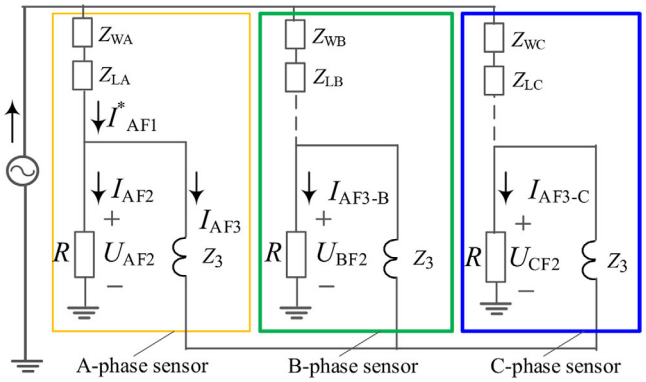


FIGURE 9 Equivalent circuit to the FRA current with respect to the A-phase sensor.

$$I_{AF2} = \frac{j\omega k n_2 z_3 I_{AF1} + j\omega k n_2 n_3 U_{F3}}{j\omega k n_2^2 z_3 + j\omega k n_3^2 R + z_3 R} \tag{21}$$

Then, let:

$$a = \frac{j\omega k n_2 z_3}{j\omega k n_2^2 z_3 + j\omega k n_3^2 R + z_3 R} \tag{22}$$

$$b = \frac{j\omega k n_2 n_3}{j\omega k n_2^2 z_3 + j\omega k n_3^2 R + z_3 R} \times \frac{j\omega k n_3 R}{3R + 3j\omega k n_2^2} \tag{23}$$

We then obtain:

$$\begin{bmatrix} U_{AF2} \\ U_{BF2} \\ U_{CF2} \end{bmatrix} = R \begin{bmatrix} I_{AF2} \\ I_{BF2} \\ I_{CF2} \end{bmatrix} = R \begin{bmatrix} a+b & b & b \\ b & a+b & b \\ b & b & a+b \end{bmatrix} \cdot \begin{bmatrix} I_{AF1} \\ I_{BF1} \\ I_{CF1} \end{bmatrix} \tag{24}$$

Then:

$$\begin{bmatrix} I_{AF1} \\ I_{BF1} \\ I_{CF1} \end{bmatrix} = \frac{1}{R} \cdot \begin{bmatrix} a+b & b & b \\ b & a+b & b \\ b & b & a+b \end{bmatrix}^{-1} \cdot \begin{bmatrix} U_{AF2} \\ U_{BF2} \\ U_{CF2} \end{bmatrix} \tag{25}$$

Let:

$$\mathbf{Z} = \begin{bmatrix} Z_{AA} & Z_{AB} & Z_{AC} \\ Z_{BA} & Z_{BB} & Z_{BC} \\ Z_{CA} & Z_{BC} & Z_{CC} \end{bmatrix} = R \begin{bmatrix} a+b & b & b \\ b & a+b & b \\ b & b & a+b \end{bmatrix} \tag{26}$$

The FRA currents I_{AF1}, I_{BF1}, and I_{CF1} can therefore be calculated by the outputs of the set of sensors provided that the impedance matrix with respect to a and b is known. Equation (25) holds only if a is not equal to zero (in other words, if z_3 is not equal to zero). The characteristics of the sensing system can be described by the impedance matrix **Z**.

To understand this phenomenon, the further simplified circuit topology in Figure 7b is shown in Figure 9, provided that RI_2 is much less than $j\omega LI_2$ and that $n_2 = n_3$. Therefore, the sensor can be considered to be an ideal transformer. In Figure 9, I^*_{AF1} is the converted value of I_{AF1} based on the number of turns of the primary-side coil and coil 2 (that is, $I^*_{AF1} = I_{AF1}/n_2$).

Our simulation-based analysis and field experiment revealed that the impedances of the windings and load of the transformer were considerably greater than the integral resistance R of the sensors in the range of frequency of the FRA. Therefore, the primary sides of the B-phase and C-phase sensors constitute an open circuit. The integral resistance R and small impedances z_3 of the B-phase sensor were parallel to those of the C-phase sensor and then were in series with the small impedance z_3 of the A-phase sensor. Therefore, the total integral impedance of sensor A is slightly less than R. Thus, the A-phase sensor still retains the required indicators (f_L and Z_C) as a single sensor for FAR measurements.

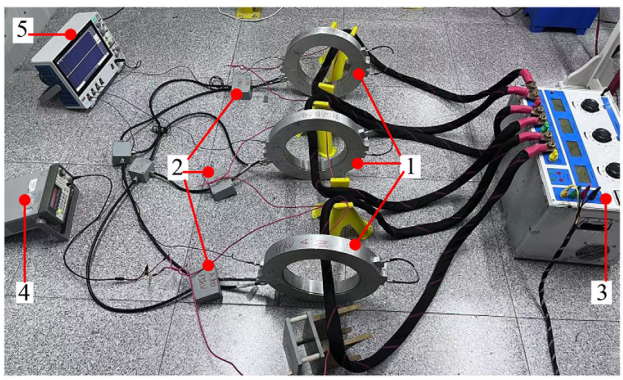
4 | EXPERIMENTAL VERIFICATION

4.1 | Setup of the verification experiment in the laboratory

In our experiments to test the proposed method, three sensors were connected to one another according to the topological structure shown in Figure 7. The details of the single sensors used are provided in Section 2. Coils 2 and 3 both exhibited 150 turns, z_3 was realized by an inductor with an inductance of 100 μH. A power frequency current generator was used to produce three-phase currents of up to 500 A. The FRA current was generated by a signal generator. The outputs of the sensors were recorded by a digital oscilloscope. This experimental setup is shown in Figure 10.

4.2 | Suppression of the power frequency magnetic field

Large, three-phase power frequency currents were supplied to the primary side of the three-phase sensors, and a high-frequency FRA current was provided to the A-phase sensor. When the amplitudes of the power frequency currents I_{AP1}, I_{BP1}, and I_{CP1} were 100 A and the amplitude and frequency of the FRA current I_{AF1} were 70.72 mA and 200 kHz, the



1-sensor; 2-inductor z_3; 3-power frequency current generator; 4-high-frequency signal generator; 5-digital oscilloscope

FIGURE 10 Experimental setup in laboratory.

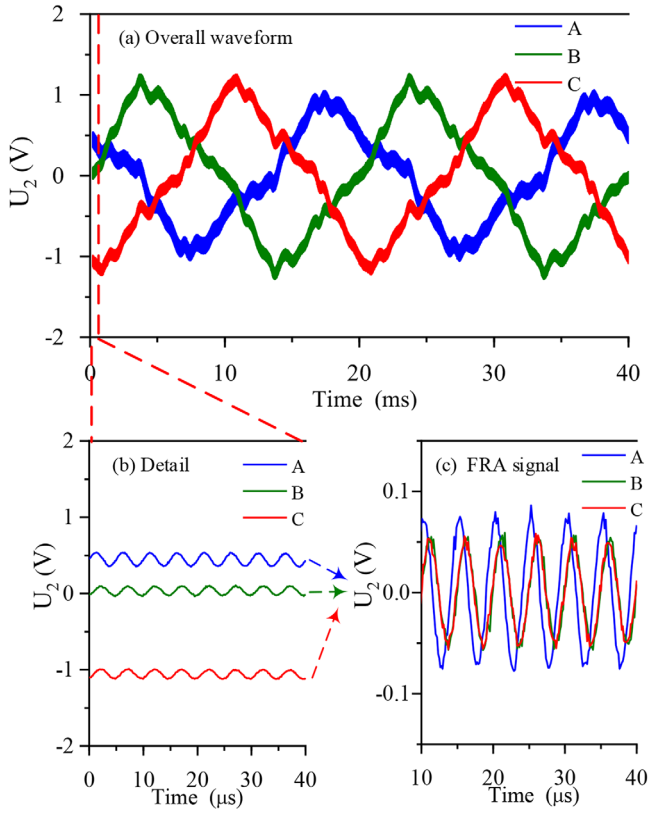


FIGURE 11 Superimposed output signals of the three-phase sensors under a 50 Hz current of 100 A and an FRA current of 200 kHz.

overall output voltages of the sensing system exhibited a power frequency waveform with an peak amplitude of approximately 1.25 V, as shown in Figure 11a. An high-resolution view of the waveform from 0 to 40 μs, as in Figure 11a, is shown in Figure 11b. This magnification makes clear the FRA signals. These FRA signals were obtained by filtering the power frequency interference in the waveforms in Figure 11b, as shown in Figure 11c.

The output voltages of the sensing system for the power frequency currents I_{AP1}, I_{BP1}, and I_{CP1} with amplitudes up to 500 A and the FRA current I_{AF1} with amplitudes and frequencies of 70.72 mA and 200 kHz are shown in Figure 12. The peak value of each of the power frequency interferences (U_{AP2}, U_{BP2}, and U_{CP2}) was approximately 6 V.

Compared to the output of the single sensor to the power frequency current shown in Figure 5, in which the single sensor has an output of 45 V to an input of 150 A, our sensing system has an output of 6 V to an input of 500 A. Clearly, our sensing system has an excellent effect on the suppression of the power frequency magnetic field in the core of the sensors.

Moreover, the amplitude of the power frequency interference in the outputs of the sensing system depends on the balance between the load currents I_{AP1}, I_{BP1}, and I_{CP1} and the amplitude of the impedance z_3. If the balance and z_3 are fixed on current value, the magnetic cores are not saturated even when the load of the transformer reaches 2333 A.

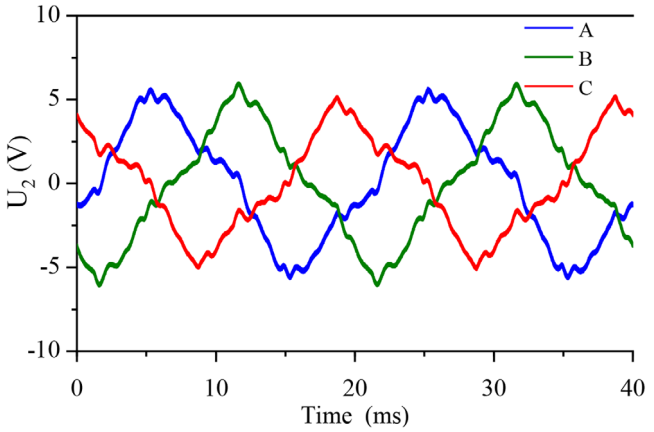


FIGURE 12 Superimposed output signals of the three-phase sensors under a 50 Hz current of 500 A and an FRA current of 200 kHz.

4.3 | Calibration of the sensing system

To distinguish the value of the FRA current on each winding of the three phases, the three-phase sensor system needed to be calibrated to obtain the parameters of the impedance matrix shown in (26). The circuit configuration shown in Figure 10 was used to carry out the calibration experiments, except that the power frequency circuit was disconnected to eliminate its influence.

The values of I_{AF1}, I_{BF1}, and I_{CF1} were set to 70.72 mA to measure U_{AF2}, U_{BF2}, and U_{CF2}, and the impedance matrix $\mathbf{Z}$ was calculated for FRA frequencies ranging from 1 kHz to 1 MHz according to (27) and (28) taking Z_{AA} and Z_{AB} as examples:

$$Z_{AA} = \left.\frac{U_{AF2}}{I_{AF1}}\right|_{I_{BF1}=I_{CF1}=0} \tag{27}$$

$$Z_{AB} = \left.\frac{U_{AF2}}{I_{BF1}}\right|_{I_{AF1}=I_{CF1}=0} \tag{28}$$

The amplitude and angle of the elements in the first line of the impedance matrix are shown in Figure 13. Corresponding values for the elements in the second and third lines were similar to those of the first line.

4.4 | Solution for the FRA response current

The accuracy of the sensing system for measuring the FRA current was verified through laboratory experiments.

A) We set the amplitude of I_{AF1} to 56.58 mA and the amplitudes of I_{BF1} and I_{CF1} to zero. The frequency ranged from 1 kHz to 1 MHz. The output voltages of U_{AF2}, U_{BF2}, and U_{CF2} were recorded. Then, we used the recorded U_{AF2}, U_{BF2}, and U_{CF2} to calculate I_{AF1}, I_{BF1} and I_{CF1} according to (25), in which the value of the impedance matrix $\mathbf{Z}$ is shown in Figure 13. The relative errors (ratios of the absolute errors to the real values of I_{AF1}) between the calculated values of

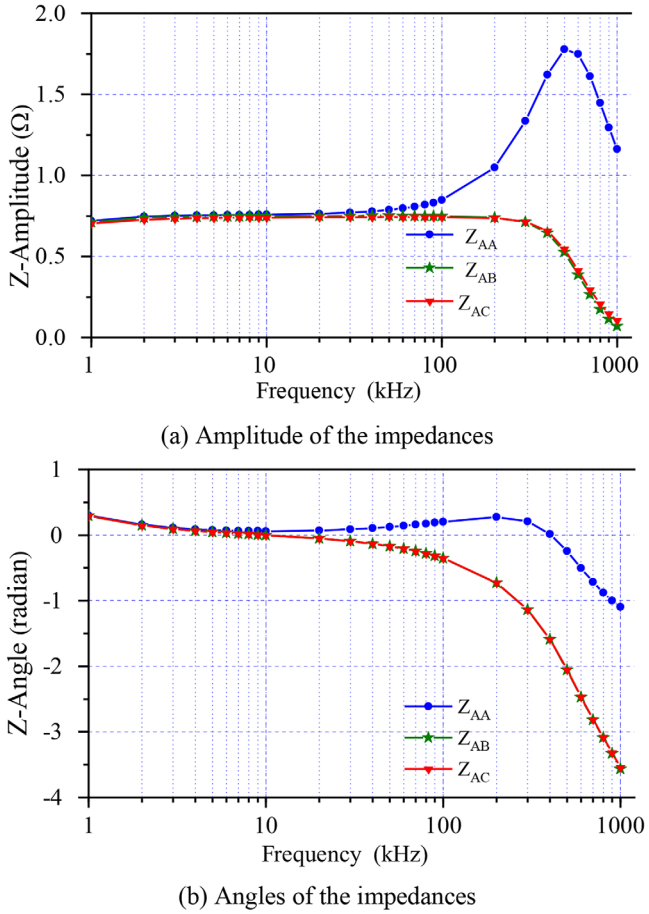


FIGURE 13 Values of the elements in the first line of the impedance matrix.

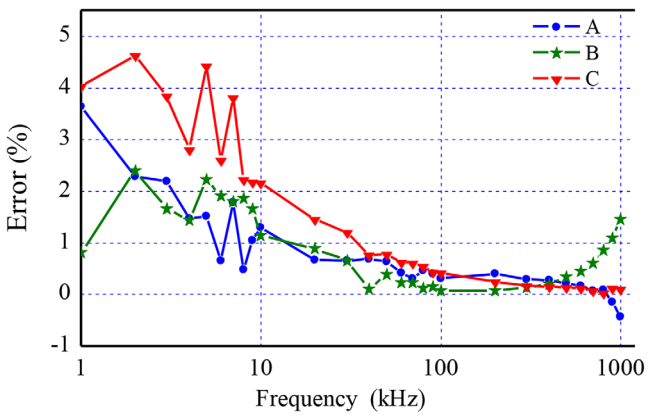


FIGURE 14 Relative errors of the three-phase sensing system in measuring the FRA current when $I_{AF1} = 56.58$ mA and $I_{BF1} = I_{CF1} = 0$.

I_{AF1}, I_{BF1}, and I_{CF1} and their true values were less than 5%, as shown in Figure 14.

B) We set I_{AF1} and I_{BF1} to the same current with an amplitude of 70.72 mA and set I_{CF1} to zero; then, we executed similar measurements and calculations. The relative errors between these calculated and true values were less than 5% in most frequency bands, as shown in Figure 15.

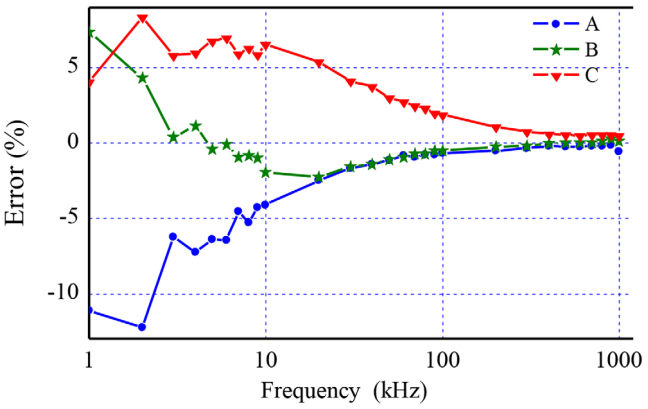


FIGURE 15 Relative error between the real FRA current and its value measured by the three-phase sensing system when $I_{AF1} = I_{BF1} = 56.58$ mA and $I_{CF1} = 0$.

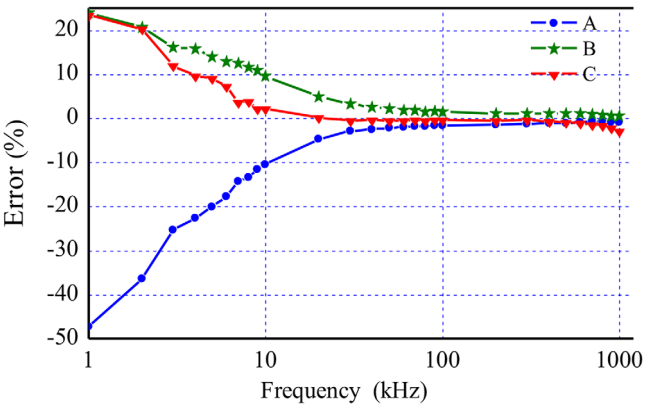


FIGURE 16 Relative error between the real FRA current and its values measured by the three-phase sensing system when $I_{AF1} = I_{BF1} = I_{CF1} = 56.58$ mA.

C) We set I_{AF1}, I_{BF1} and I_{CF1} to the same current with an amplitude of 70.72 mA and executed similar measurements and calculations. The relative errors were still smaller than 5% in the major frequency bands, as shown in Figure 16. The overall relative error of the sensing system was less than 2% in the range of frequencies from 60 to 800 kHz.

4.5 | Experimental results on the transformer

Experiments were finally executed on an SSZ11-50000/110 type of transformer. The transformer and the sensor installation were shown in Figure 2. While conducting the test, the transformer was not connected to the power grid of its substation. Therefore, the high voltage terminals of each winding were connected to ground through 50 Ω resistors. That is, the power grid was temporary represented by the resistors.

The neutral points of the windings with voltages of 35 and 110 kV were connected to the ground, while the excitation coil was installed on the grounding line of the neutral point, as shown in Figure 2a. The FRA current sensors were installed at the boots of the bushings of the 35 kV windings, as shown in

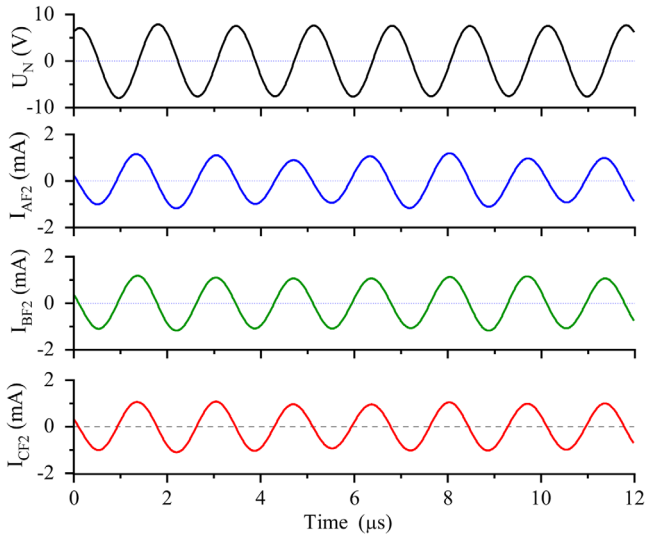


FIGURE 17 Current measurements for FRA (600 kHz) on 35 kV windings of the power transformer.

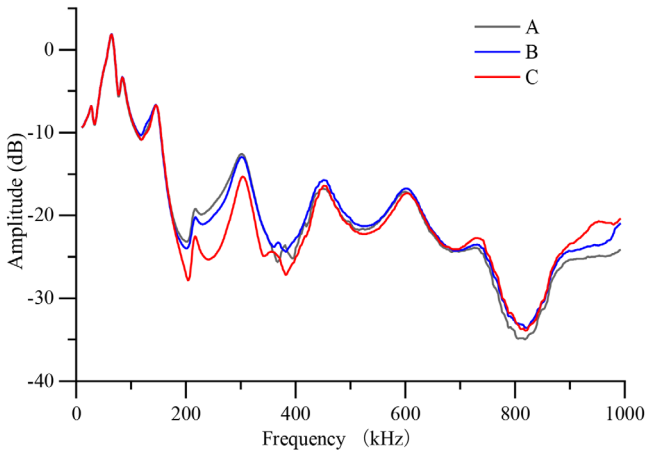


FIGURE 18 The amplitude–frequency curve of the transfer function of the current of the 35 kV windings of the power transformer.

Figure 2b. The sensing system was constructed as shown in the circuit in Figure 7.

A voltage source was used to generate a U_S, with a sweeping frequency of a sine signal and an amplitude of 100 V. When U_S was applied to the excitation coil, a potential of approximately 5 V was induced at the neutral point. Some of the output waveforms of the sensing system on the 35 kV windings are shown in Figure 17. Because the excitation signal was injected into the neutral point, the FRA currents on the terminals of the three windings were basically the same. These results were consistent with our previous theoretical analysis.

Based on the FRA currents, we propose a current transfer function for the transformer. This function describes the ratio of the FRA current on the high voltage terminal of the winding to the excitation current on the neutral point. The amplitude-frequency curve of the transfer function is shown in Figure 18. Although this curve differs from the previous curve (with a function described by the ratio of the FRA voltage to the excitation voltage), it also reflects the resonance phenomenon in the windings. Therefore, the transfer function of the current can be used in the diagnosis of winding deformation.

5 | DISCUSSION

The function of z_3 is displayed in (22)–(23). If $\mathcal{Z}_3 = 0$, then $a = 0$, the element in the impedance matrix exhibits the same value, and it has no inverse matrix. Therefore, the FRA current on each winding cannot be determined by the outputs of the sensing system. Thus, the function z_3 ensures that the matrix is invertible. In other words, this configuration ensures that the sensor in one phase has different outputs from the FRA response current in the same phase as well as that in another phase. For example, U_{AF2} exhibits different values to I_{AF1} and I_{BF1}. $\mathcal{Z}_3$ should be set to be sufficiently large to ensure this difference. Therefore, the sensing system can distinguish among I_{AF1}, I_{BF1}, and I_{CF1}. $\mathcal{Z}_3$ was set to 100 μH in our experiments. The difference in impedance among the three phases was minimal at frequencies below 60 kHz. Therefore, the relative error in the range of 1 to 60 kHz was larger than that in the range of 60 kHz to 1 MHz as shown in Figures 14–16.

However, $\mathcal{Z}_3$ determines the amplitude of the power frequency interference in the output of the sensing system and the magnetic flux of the power frequency B_{mag} in the core of the sensors. As shown in Figure 8, U_{AP2} increases with increasing impedance of z_3. To reduce U_{AP2} and B_{mag} as much as possible, z_3 should be set to a very small value.

In sum, z_3 should be sufficiently small at the given power frequency and sufficiently large in the range of frequency of the FRA. An inductor is a good choice for z_3.

6 | CONCLUSION

We present a method to suppress the power frequency magnetic field in the cores of sensors for FRA on live transformers based on the basic phenomenon that the sum of the three-phase power frequency load currents is close to zero. The proposed method consists of a sensing system with a special connection among three-phase current sensors. Unlike conventional methods, this sensing system can drastically suppress the power frequency magnetic field in the cores of the sensors under large load currents of the transformer while maintaining the frequency band required for FRA.

Compared to a single sensor, which has an output of 45 V to an input of 150 A, our sensing system has an output of 6 V to an input of 500 A. The power frequency magnetic field in the core of the sensor and the power frequency interference in the output of the sensors were significantly compressed.

The FRA currents in the three phase windings can be measured and distinguished by using the outputs of the three-phase sensing system. The relative error was found to be less than 2% in the range of frequencies from 60 to 800 kHz.

AUTHOR CONTRIBUTIONS

Contributed the conceptualization, methodology, data process, and writing: Yangchun Cheng. *Contributed the sensing system manufacture, experiment in laboratory and data process*: Xiangdong Liu. *Contributed the experiment in laboratory*: Yufei Sha. Contributed the experiment in power station and project management: Wenzhi Chang. *Contributed the funding support and project management*: Jiangang Bi.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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